



Incident handler's journal

Author: Cybersecurity Portfolio Document

Framework Alignment: NIST Incident Response Lifecycle (SP 800-61 Rev. 2)

Repository Purpose: Portfolio demonstration of Incident Response, Threat Investigation, Log Analysis, and SIEM/IDS Tooling

Executive Summary

This document serves as a professional Incident Handler's Journal documenting incident investigations, security log analysis, threat hunting, and detection engineering. Each entry maps directly to specific phases of the NIST Incident Response Lifecycle: *Preparation, Detection & Analysis, Containment, Eradication, & Recovery, or Post-Incident Activity.*

Date: June 14, 2026	Entry: 02 Investigation — Suspicious Authentication Anomaly & Lateral Movement
Description	Detection and Analysis: The scenario outlines how the organization first detected unauthorized authentication anomalies using Splunk SIEM. For the analysis step, the organization analyzed authentication logs and correlated event codes to determine the scope of compromised accounts. Containment, Eradication, and Recovery: The scenario details some steps that the organization took to contain the incident. For example, the company temporarily locked affected domain accounts and updated perimeter firewall policies. However, to eradicate the threat and safely recover services, they coordinated with specialized internal

	teams to enforce multi-factor authentication across all endpoints.
Tool(s) used	Splunk SIEM
The 5 W's	Capture the 5 W's of an incident. • Who: An unknown external threat actor utilizing an automated brute-force script. caused the incident? • What: Multiple failed authentication attempts followed by a successful login outside standard business hours. • When: June 14, 2026, between 02:15 UTC and 03:45 UTC • Where: Internal domain controller and single sign-on (SSO) authentication server. • Why: To execute password spraying, achieve initial access, and attempt lateral movement across host systems.
Additional notes	Executed targeted Search Processing Language (SPL) queries to filter event codes (4625 failed logins, 4624 successful logins)

Date: July 08, 2026 .	Entry: 02 Investigation — Malicious Process Execution & Endpoint Isolation
Description	This investigation unfolded across two distinct operational phases: 1. Detection and Analysis: An automated endpoint alert flagged an unverified executable attempting outbound communication to an unknown external domain. Analysts evaluated process trees and network connections to verify the severity of the execution. 2. Containment, Eradication, and Recovery: The host was immediately severed from the local network via endpoint containment controls. The

	suspicious binary was quarantined, malicious registry modifications were removed, and the host image was validated before restoring network access
Tool(s) used	Google Chronicle
The 5 W's	Capture the 5 W's of an incident. • Who: Unknown malicious actor targeting endpoint users through web-based staging vectors. • What: Execution of a secondary payload establishing C2 callback channels. • When: July 07, 2026, at 11:30 UTC. • Where: Finance Department Workstation (FIN-WS-08). • Why: To establish persistent administrative access and stage secondary tools.
Additional notes	Constructed Universal Data Model (UDM) queries in Google Chronicle to trace parent-child process lineages (target.process.file_name) and map outbound socket connections.

Date: August 19, 2026	Entry: 03 Engineering — Custom Detection Signatures for Threat Probing
Description	Engineered and deployed custom Intrusion Detection System (IDS) signature rules to spot unauthorized network reconnaissance and policy violations. This task maps directly to the Preparation phase of the NIST Incident Response Lifecycle by sharpening detection optics prior to security incidents.
Tool(s) used	Suricata IDS .

The 5 W's	Capture the 5 W's of an incident. • Who: N/A (Detection Engineering & Rule Authoring) • What: N/A (Detection Engineering & Rule Authoring) • When: N/A (Detection Engineering & Rule Authoring) • Where: N/A (Detection Engineering & Rule Authoring) • Why: N/A (Detection Engineering & Rule Authoring)
Additional notes	Authored custom Suricata rules specifying action headers, protocol matchers, IP variables, and explicit rule options (content, sid, rev). Validated rules against synthetic PCAP traffic to confirm alerting efficiency while preventing rule-trigger noise.

Date: September 08, 2026	Entry: 04 Threat Hunting — Telemetry Correlation Across Enterprise Systems
Description	Executed a proactive threat-hunting campaign aggregating web proxy, DNS, and endpoint telemetry to catch stealthy beaconing activity. This initiative falls under the Detection & Analysis phase of the NIST IR framework.
Tool(s) used	Google Chronicle & Splunk SIEM
The 5 W's	Capture the 5 W's of an incident. • Who: N/A (Proactive Threat Hunting & Analytics) • What: N/A (Proactive Threat Hunting & Analytics) • When: N/A (Proactive Threat Hunting & Analytics) • Where: N/A (Proactive Threat Hunting & Analytics) • Why: N/A (Proactive Threat Hunting & Analytics)

Additional notes	Correlated high-volume UDM telemetry in Chronicle alongside Splunk dashboard panels to detect anomalous HTTP user-agent strings and periodic domain polling, helping validate overall network perimeter posture.
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Reflections/Notes:

Personal Reflection & Engineering Notes

1. Were there any specific activities that were challenging for you? Why or why not?

Getting comfortable with query syntax—specifically crafting efficient SPL searches in Splunk and exact signature logic in Suricata—was the most challenging aspect. Early on, broad queries returned far too much benign log noise. Learning how to apply precise logical operators, field matchers, and wildcards took practice, but it significantly speeded up my ability to isolate relevant indicators.

2. Has your understanding of incident detection and response changed since taking this course?

My perspective shifted from viewing incident response as a series of ad-hoc cleanup steps to recognizing it as a structured lifecycle. I used to think security work was mostly about fixing compromised hosts after an event. Now I understand that continuous logging, disciplined documentation, clear communication, and standardized frameworks (like NIST) are what actually make a defense resilient.

3. Was there a specific tool or concept that you enjoyed the most? Why?

Working with SIEM platforms like Splunk and Google Chronicle was definitely the highlight. There's something really satisfying about taking millions of unorganized, raw system events and refining them into a clear narrative of what happened using targeted search queries. It gives you complete visibility across the whole network environment.